Affinity Drifting: Learning Conditional Distributions from One Sample per Condition

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Working draft — results update automatically as experiments finish (2026-11-08 13:49 UTC, commit `75d2109`)

Abstract

Drifting models train a one-step generator by moving its samples along a field that attracts them to data and repels them from each other. In text-to-image data, however, each caption usually comes with a single image. We show that with one positive per condition the attraction term reduces exactly to L_2 regression onto that positive, so a generator that cannot resolve nearby conditions collapses to the conditional mean. We propose Affinity Drifting (AffD), which replaces the single target by a kernel mean-shift over the positives of semantically similar conditions, weighted by a caption-space affinity. For Gaussian kernels the field is the gradient of a log density ratio, which lets us predict its fixed points: borrowing positives with self-only repulsion converges to a Nadaraya–Watson mixture, while symmetric repulsion memorises. We test this bottom-up with pre-registered, single-factor experiments. In a non-parametric particle system the predicted fixed points are reached at the resolution of the kernel. In a 2-D toy whose conditional distributions are known, the original Drifting objective and the single-positive objective both collapse to the conditional mean (mean-collapse index 1 $[1, 1]$ and 0.9996 $[0.9995, 0.9997]$), while an oracle with many positives does not (0.09065 $[0.08467, 0.09446]$). Borrowing positives restores the mixture: AffD covers every mode and lowers the held-out W_2 to 0.33 of the single-positive objective, matching the oracle. Controls show why: averaging the borrowed targets without the x-dependent kernel still collapses, and random instead of semantic neighbours spread samples off the conditional manifold.

1. Introduction

One-step generators trained by *drifting* [2] evolve the push-forward of a noise distribution during training rather than at sampling time: each generated sample is attracted toward data and repelled from the other samples, and the generator regresses onto its own drifted outputs. With class labels, every class has many images and the attraction is a proper mean-shift toward the class distribution.

Text conditioning breaks that assumption. A caption typically describes one image, so the positive set of a condition has size one. Then the normalised attraction weight is one and the attraction no longer depends on where the sample is: it is L_2 regression onto the single positive (Lemma 1). A generator with finite resolution in c averages the positives of conditions it cannot tell apart and outputs their mean — a blurred blob between modes (Fig. 1).

We ask whether *borrowing* positives from semantically close conditions restores a distribution, and why. AffD weighs every positive y_i by a caption affinity $w_i(c)$ and keeps the x-dependent kernel of drifting, so the attraction becomes a mean-shift toward a mixture rather than a pull toward an average. The claim is mechanistic: the gain should come from the x-dependent mode assignment, not merely from having more targets. We therefore test it with controls that remove each alternative explanation in turn — averaging targets without x-dependence (K1), random instead of semantic neighbours (K2), noise-smoothed single targets (K3) — against an oracle with true samples (K4) and the original Drifting objective as a baseline.

Contributions. (i) The single-positive degeneracy of drifting and its consequence, mean collapse under limited conditional capacity. (ii) AffD, with a fixed-point analysis for Gaussian kernels. (iii) A pre-registered, single-factor experimental protocol, run bottom-up from particles to neural toys; this draft reports what is finished and marks the rest as pending.

2. Method

Drifting. A generator $f_\theta(\epsilon, c)$ maps noise to samples. Given a field $V(x \mid c)$, it is trained with $\mathcal{L} = \|f_\theta(\epsilon, c) - \mathrm{sg}(f_\theta(\epsilon, c) + V)\|^2$, where sg stops the gradient. The field is attraction minus repulsion, $V = V^+ - V^-$, each a kernel mean-shift:

$$V^+(x \mid c) = \frac{\sum_i w_i(c)\, k(x, y_i)\, (y_i - x)}{\sum_i w_i(c)\, k(x, y_i)}, \qquad (1)$$

and V^- the same over the other generated samples. In the original method the positives are the samples of the condi-

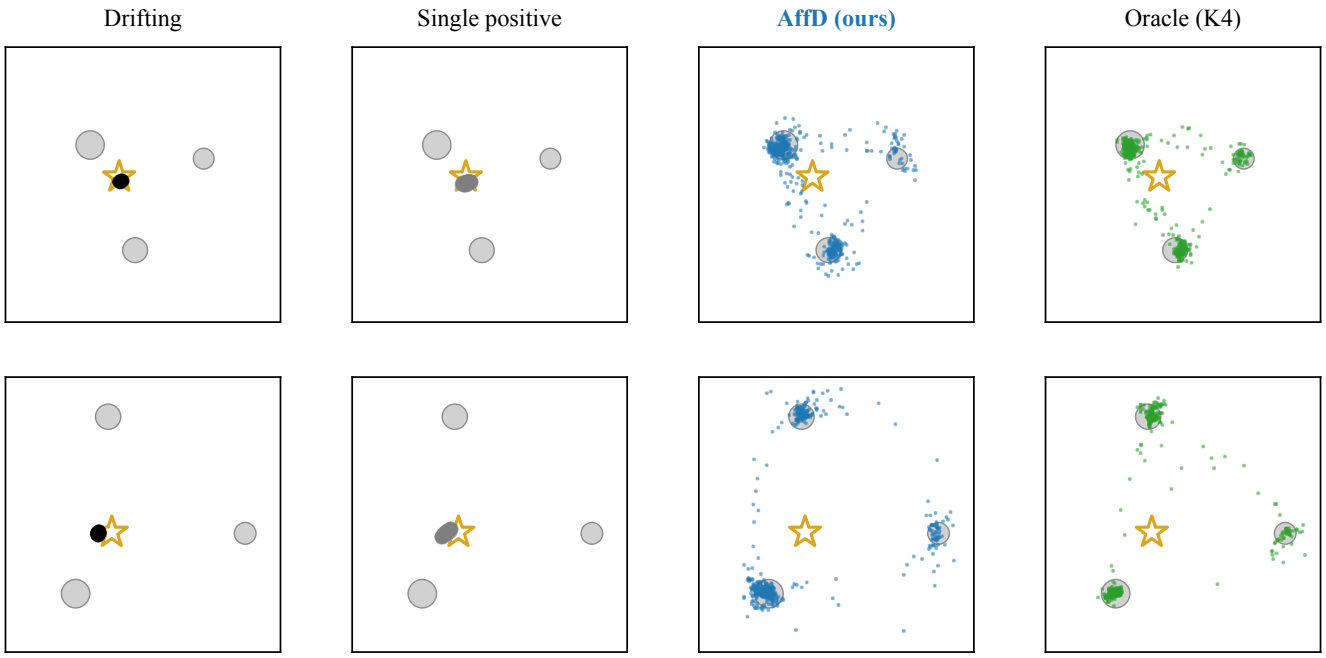


Figure 1. **Mean collapse with one positive per condition.** Samples (512 per panel) from generators trained on a 2-D toy where each training condition c has exactly one sample, on two held-out conditions. Grey: true $p(x \mid c)$, a three-mode mixture; hollow $\star$: the conditional mean $\mathbb{E}[x \mid c]$. The original Drifting objective and the single-positive objective put every sample at the conditional mean, between the modes; an oracle given many true samples per condition covers the modes. AffD uses only the single positives, borrowed across similar conditions, and recovers the modes. Held-out conditions, seed 0 of five.

tion's own class, $w_i(c) = \mathbf{1}[c_i = c]$, and the field is normalised to unit energy per batch.

Lemma 1 (Single positive is regression) *If condition c has one positive y and repulsion is off, $V^+(x \mid c) = y - x$ for every kernel, so $\nabla_\theta \mathcal{L} = \nabla_\theta \|f_\theta - y\|^2$.*

Affinity Drifting. AffD keeps (1) but borrows positives from similar conditions,

$$w_i(c) = \exp\big(-\|\psi(c) - \psi(c_i)\|^2 / 2h_p(c)^2\big), \qquad (2)$$

with a frozen condition embedding ψ (an oracle in the toy, a text encoder later). The bandwidth $h_p(c)$ is the smallest value whose effective number of neighbours $N_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ reaches $N^* = 16$, so every condition borrows equally strongly. Repulsion uses a caption bandwidth h_q; by default $h_q = 0$, *i.e.* a sample is repelled only by the other samples of its own condition. The single-positive objective is the limit $h_p \to 0$ of the same code; unconditional drifting is $h_p, h_q \to \infty$.

Fixed points. For a Gaussian kernel of width σ, (1) is a scaled score:

Lemma 2 (Score ratio) *$V(x \mid c) = \sigma^2 \nabla_x \big[\log \hat{p}(x \mid c) - \log \hat{q}(x \mid c)\big]$, where $\hat{p} = \sum_i w_i(c)\, k_\sigma(\cdot, y_i)$ and $\hat{q}$ is the kernel density of the generated samples.*

Hence $V \equiv 0$ exactly when $\hat{q} = \hat{p}$. With $h_q = 0$ the equilibrium of a condition is the Nadaraya–Watson mixture of its borrowed positives, $q_c = \sum_i \tilde{w}_i(c)\, \delta_{y_i}$ with $\tilde{w} = w / \sum w$ (Proposition C in our plan); with symmetric repulsion $h_q = h_p$ the mixing cancels on both sides and every condition memorises its own positive (Proposition A). The bandwidth h_q thus selects the equilibrium.

Implementation choices. We use a mixture of kernel widths, $\sigma \in \{0.071, 0.384, 2.07\}$ from the N^*-th-nearest-neighbour distance to the median pairwise distance of the positives, summing one field per width; each width satisfies Lemma 2, so the fixed point is unchanged. As in the original method, the *whole* field is normalised to unit energy per batch and per width. A per-term normalisation, which we first used, keeps V^+ and V^- at unit size even where both should vanish and flung a few percent of single-positive samples far outside the data; we report it in Sec. 4.

3. Experiments

Protocol. Every comparison changes one knob of a base configuration; a run refuses to start if its configuration differs from the declared change. Predictions and falsification criteria are committed before any run, and verdicts (pass / fail / inconclusive) are computed from those committed

rules only. All arms share seeds (five per experiment), probe inputs and the learning rate, which is fixed once on an unconditional run. Intervals are 95% paired bootstrap CIs over seeds. Every experiment also trains the original Drifting objective, unchanged, as a reference that is reported but not judged.

3.1. Fixed points in a particle system

Without a network, each of 32 conditions on a line owns 256 particles moved by $x \leftarrow x + \eta V(x \mid c)$, *i.e.* a generator of unlimited capacity, with one positive per condition. The field matches the autograd score ratio of Lemma 2 to 1.6×10^{-15} (float64; PASS). Increasing h_q from 0 to h_p moves the equilibrium monotonically from the mixture to memorisation (Fig. 2; PASS), whereas the original Drifting objective memorises outright (own-positive share 0.951).

The point-level assignment, however, misses the pre-registered tolerance for the mixture (TV 0.147 [0.145, 0.149] against 0.1; FAIL): with a kernel wider than the spacing of the positives, the field cannot resolve which nearby positive a particle sits on. Compared at the kernel's resolution — TV between the σ-smoothed densities, an analysis we did not pre-register — the equilibrium matches $\hat{p}$ to 1.0×10^{-4}. A counterfactual negative between two conditions has the predicted closed-form Gaussian equilibrium in its algebra (relative error 3.3×10^{-6} under a Gaussian closure), but with finite samples it is stable only for small weights (error 0.0278 at $\lambda = 0.25$, divergence from $\lambda = 0.5$; FAIL).

3.2. Mean collapse and its repair in a 2-D toy

Setup. Conditions $c \in [0, 1]^2$ are drawn once; each of the 10,000 training conditions has a single positive from $p(x \mid c)$, a three-mode Gaussian mixture whose modes rotate with c_1 and whose radius and weights change with c_2, so the conditional mean falls between modes. An MLP generator $f(\epsilon, g(c))$ sees c quantised to cells of width $\delta = 1/16$, roughly 39 training conditions per cell. We train with Adam (10^{-3}, chosen on the unconditional model: marginal W_2 0.136 [0.129, 0.144] against a sampling floor of 0.0876), an EMA of the weights, 20,000 steps, and 8 conditions $\times$ 64 samples per batch, and evaluate on 256 held-out conditions with 512 samples each. The mean-collapse index is $\mathrm{MCI} = 1 - \operatorname{tr}\mathrm{Cov}_q / \operatorname{tr}\mathrm{Cov}_p$.

Single positive collapses (C1). Both the original Drifting objective and the single-positive objective collapse every condition to a point at the conditional mean (MCI 1 [1, 1] and 0.9996 [0.9995, 0.9997]; off-manifold share 1), while the oracle with 16 true samples per condition and step covers the modes (MCI 0.09065 [0.08467, 0.09446], mode coverage 0.977). This is the failure AffD targets (PASS).

Method	W2 ↓	MCI ↓	Off-man. ↓	Mode cov. ↑	Recall ↑
Drifting	1.4	1	1	0.34	0
Single positive	1.4	1	1	0.35	1.5×10^{-6}
K1 ($k_x \equiv 1$)	1.3	0.99	1	0.48	0.007
K2 (random nbrs)	1.1	−0.29	0.92	1	0.97
K3 (noisy single)	1.4	1	1	0.36	0
AffD (ours)	0.47	0.026	0.28	1	0.94
Oracle (K4)	0.5	0.091	0.18	0.98	0.78

Table 1. **2-D toy, held-out conditions** (seed means). Above the line: the single-positive regime and the three mechanism controls; below: AffD and the oracle upper bound, which sees 16 true samples per condition and step.

Borrowing restores the mixture (C2). AffD, trained on the same single positives, covers every mode (mode coverage 1, recall 0.94) and reaches a held-out W_2 of 0.469 [0.46, 0.477] against 1.41 [1.41, 1.42] for the single-positive objective (ratio 0.33; Tab. 1), on par with the oracle that sees true samples. The pre-registered comparison is PASS: AffD beats the single-positive objective and each control with paired CIs excluding zero. The controls fail in the ways the mechanism predicts. K1 borrows the same positives but drops the x-dependent kernel, so its target is their weighted average — and it collapses like the single positive (MCI 0.9878). K2 borrows as many positives at random: samples spread over all modes (coverage 1) but most fall off the conditional manifold (off-manifold share 0.922). Noise-smoothed single targets (K3) still collapse (MCI 0.9996). The gain therefore comes from the x-dependent mode assignment of semantically close positives, not from more targets or smoother ones. Figure 3 shows that assignment: with one positive every attraction arrow points at the same target, whereas the borrowed positives split the space into one basin per mode. One caveat is visible already: with ample capacity, a share of AffD's samples lands on borrowed training positives (memorisation share 0.165 on training conditions, against 0.00443 for the single positive), as the fixed-point analysis predicts.

The gap grows with lost capacity (C3). [pending] (Fig. 4).

4. Status, deviations and limitations

This is a working draft; Tab. 2 lists every pre-registered experiment and its verdict at the time of the build. Changes to the plan made along the way are logged with reasons and approvals; the ones that matter for reading the results are: the learning-rate gate failed twice before passing, first because unconditional drifting was mis-specified (repulsion within each condition's samples instead of the whole batch) and then narrowly; a single-width kernel transported mass too locally and was replaced by a mixture of widths; 64 instead

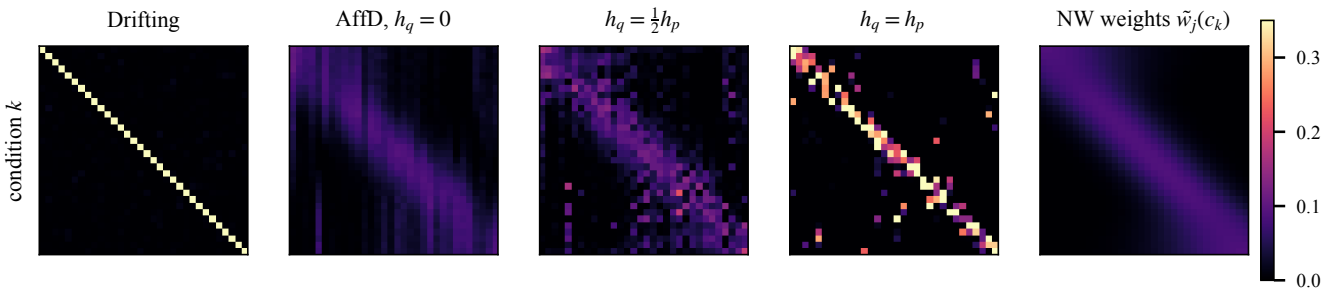


Figure 2. h_q **selects the equilibrium** (particles, seed mean). Share of each condition's particles nearest to each positive. Original Drifting memorises (diagonal). AffD with $h_q = 0$ spreads mass over the borrowed neighbours, as the Nadaraya–Watson weights $\tilde{w}$ predict (right); larger h_q narrows the band toward memorisation.

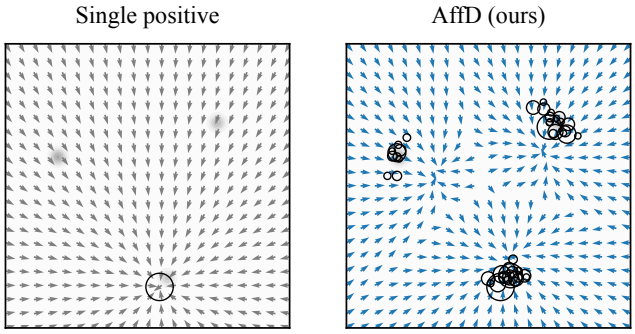


Figure 3. **Mechanism.** Direction of the attraction $V^+(x \mid c)$ for one training condition. Single positive: every arrow points at the one target. AffD: arrows point to the nearest mode of the borrowed positives (circles, size $\propto$ weight).



Figure 4. **Capacity.** W_2 of the single-positive objective and of AffD as the generator's resolution in c coarsens (seed mean, 95% CI).

of 8 samples per condition are generated, since self-only repulsion estimated from 8 samples is strongly biased; and the drift normalisation was corrected from per-term to whole-field, as in the original method, after the per-term version flung single-positive samples outside the data. Runs made before a correction are kept and excluded.

The toy uses an oracle condition embedding; text embeddings, learned features and real captions are later stages of the plan, not reported here.

Level	ID	Claim	Verdict
L0	E0.1	Lemma 1	PASS
L0	E0.2	Prop A	INCONCLUSIVE
L0	E0.3	Prop C	FAIL
L0	E0.4	Prop A–C	PASS
L0	E0.5	Prop E	FAIL
L0	E0.5b	Prop E (diagnostic)	PASS
L0	E0.6	Normalisation (cue 1)	FAIL
L1	E1.1	C0	PASS
L1	E1.2	C1	PASS
L1	E1.3	C2	PASS
L1	E1.4b	C3	[pending]
L1	E1.4	C3	[pending]
L1	E1.5	C4	[pending]
L1	E1.6	Bandwidth selection	[pending]
L1	G4	Gate G4	FAIL
L1	G4b	Gate G4 (2nd)	FAIL
L1	G4c	Gate G4 (3rd)	PASS

Table 2. Pre-registered experiments and their verdicts at build time.

5. Related work

AffD builds on drifting [2], which turns kernel mean-shift [1] into a training signal for one-step generators. Its attraction is a Nadaraya–Watson kernel regression [7, 9] over borrowed positives, and the field is closely related to gradients of kernel two-sample statistics (MMD) [3], which generative models have used with mixtures of kernel widths for the same reason we do [5]. Our evaluation follows standard practice for mode collapse: precision and recall [4] and density and coverage [8]. The capacity control by spectral normalisation follows [6].

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